

1 Explaining Technological Progress: economics of ideas and growth

The Solow model shows that capital accumulation cannot generate sustained growth in per capita terms. As the capital stock increases, output rises at a diminishing rate, and eventually these gains are offset by depreciation, leading the economy to converge to a steady state with no per capita growth. Sustained growth in per capita income must therefore come from productivity growth. However, in the Solow model, productivity growth is exogenous—the model does not explain how it is determined.

A major step toward explaining the determination of productivity growth was made by Paul Romer (1990), whose work earned him the Nobel Prize in 2018. His key insight is that productivity growth is driven by the creation of new ideas. The production of ideas depends on the number of individuals engaged in research, which in turn is determined by the incentives to innovate—namely, the profits that successful innovators expect to earn. Ideas differ in a fundamental way from capital: while the use of a unit of capital by one firm prevents its use by another, this is not the case for ideas. The use of an idea by one agent does not prevent others from using it as well; that is, ideas are non-rival. Once created, they can be used by many firms simultaneously without being depleted. At the same time, ideas are only partially excludable. In the absence of legal protection, it is often difficult for innovators to prevent others from copying and using their discoveries, thereby eroding post-innovation profits and discouraging innovation in the first place (why would a private firm incur the fixed costs of research if the resulting innovation were effectively a public good, yielding zero profits?). Institutions such as patents seek to address this problem by granting innovators temporary monopoly rights over the use of their ideas, thereby allowing them to earn profits and recover their research costs. However, because these rights are imperfect—knowledge can often diffuse despite legal protections—excludability is never complete.

Technology can be broadly defined as the process by which inputs are transformed into output, as represented by the production function. The accumulation of knowledge expands the set of available production technologies. A new idea may allow a given set of inputs to produce more output or higher-quality output—a process known as vertical innovation—potentially rendering existing technologies obsolete. Alternatively, a new idea may introduce a new product, expanding the set of available goods—this is horizontal innovation. Romer’s model of expanding varieties focuses on horizontal innovation. Vertical innovation is central to Schumpeterian growth models, such as those developed by Philippe Aghion and Peter Howitt (1992), which earned them the Nobel Prize in 2025. In these models, growth is the result of a process of creative destruction, in which new innovations displace old ones,

generating both winners and losers.

These models allow us to address a normative question: is the amount of research optimal? Firms undertake research to maximize their flow of profits, that is, for private incentives, which may not be aligned with the social incentives of innovation.

On the one hand, innovation generates knowledge spillovers that are not internalized by the innovators. New ideas build on existing ones and facilitate future innovations, often by different agents. This idea is captured in the famous quote by Newton: "If I have seen further, it is by standing on the shoulders of giants". As a result, the social returns to innovation may exceed the private returns, suggesting that the equilibrium level of research may be inefficiently low and providing a rationale for research subsidies.

On the other hand, Schumpeterian models also highlight the possibility of excessive research. Because innovation displaces existing firms, it generates losses that innovators do not internalize. Moreover, private gains may exceed social gains when innovations yield only small improvements in quality or productivity but allow the innovator to capture a large share of the market. For example, a firm that introduces a marginal improvement in headphone sound quality may generate limited benefits for consumers, yet still earn large profits by displacing incumbent producers.

1.1 Economics of Ideas

Ideas are non-rival (I can use an idea from calculus at the same time as you) and generally non-excludable (I cannot easily prevent you from using it).

The fact that ideas are non-rival implies that they only need to be produced once. Ideas require an initial effort (a fixed cost) to be created—for example, inventing a vaccine or producing a movie—but once created, they have low (or near-zero) marginal cost (e.g., the cost of an additional dose of a vaccine or of streaming the movie to one more user). The presence of fixed costs implies increasing returns to scale. To see this, consider the following production function: given labor L , output is

$$y(L) = 10(L - F),$$

where F is the amount of labor devoted to fixed costs. If we double labor, output more than doubles:

$$y(2L) = 10(2L - F) > 10(2L - 2F) = 2y(L).$$

Increasing returns to scale imply that average costs exceed marginal costs. As a result, ideas will only be produced if firms can charge prices above marginal cost; that is, some

form of imperfect competition is required. Such profits can be sustained only if others are prevented from freely using the idea, through mechanisms such as patents, copyright, branding, or secrecy.

1.2 A toy version of the expanding variety model (Romer, 1990)

We will assume:

1. We simplify the household problem: as in the Solow model, we assume that households save a constant fraction s of income Y_t ; thus, aggregate savings are given by $S_t = s, Y_t$.
2. The population N_t grows at a constant rate: $N_{t+1} = n N_t$. N_t is also the total labor supplied.
3. The aggregate production function is given by $Y_t = K_t^\alpha (A_t L_t)^{1-\alpha}$, where K_t is the capital stock in period t , A_t is a productivity variable that (as defined below) grows with the stock of ideas, and L_t is the amount of labor used in production in period t .
4. A fraction δ of capital depreciates each period; thus, the law of motion of capital is (as in the Solow model): $K_{t+1} = s Y_t + (1 - \delta) K_t$.
5. The production function for new ideas is given by:

$$A_{t+1} - A_t = \theta L_{R,t}^\lambda A_t^\phi,$$

where θ is a scale parameter that captures how the flow of ideas translates into changes in productivity, $L_{R,t}$ is the amount of labor employed in research (researchers), $\lambda \in (0, 1)$ captures diminishing returns to researchers, and ϕ captures knowledge spillovers from the existing stock of ideas.

- $\phi \in (0, 1)$: positive knowledge spillovers, where existing knowledge facilitates the generation of new ideas (“standing on the shoulders of giants”). As will become evident later, the restriction $\phi < 1$ is imposed to guarantee the existence of a Balanced Growth Path. We say that models with $\phi < 1$ exhibit *weak* scale effects because s_R does not affect growth rates along the BGP (as we will see below, the growth rate of A_t in the BGP depends only on population growth). By contrast, models with *strong* scale effects predict that a higher s_R leads to higher growth in the BGP; however, the empirical evidence is not consistent with this prediction, as growth has remained relatively stable over time despite a substantial increase in research effort.

- $\phi < 0$: this captures the idea that ideas are becoming harder to find; that is, maintaining the same flow of new ideas requires the number of researchers to grow over time.
 - $\phi = 0$: the flow of new ideas is independent of the current stock of knowledge.
6. We also simplify the allocation of labor: an exogenous constant fraction s_R of the total labor force is allocated to research; thus, $L_{R,t} = s_R N_t$ and $L_t = (1 - s_R) N_t$. Note that this implies a trade-off for social welfare: a higher fraction of labor allocated to research allows for higher production in the future, but at the cost of lower production today (as it implies less labor used in production).

This toy model only partially captures Romer's contribution: it delivers that growth is driven by innovation carried out by researchers, but it remains silent on another key aspect emphasized by Romer, namely that the amount of resources devoted to research is driven by the rents that innovators expect to obtain from successful innovation. In the Appendix (which you do not need to read), I present a version of the model in which the allocation of labor to research is the outcome of an optimal decision; this version shows that higher expected profits lead to a higher fraction of labor allocated to research.

Note that there are no choices, so solving the model is very mechanical:

1. We start with some initial levels K_0, N_0, A_0 .
2. Iterating over time (starting with $t = 0$), given K_t, N_t, A_t :
 - (a) Compute $Y_t = K_t^\alpha (A_t (1 - s_R) N_t)^{1-\alpha}$;
 - (b) Compute $K_{t+1} = s Y_t + (1 - \delta) K_t$;
 - (c) Compute $A_{t+1} = A_t \left(1 + \theta s_R^\lambda N_t^\lambda A_t^{\phi-1}\right)$;
 - (d) Compute $N_{t+1} = n N_t$.

Next, we ask whether a balanced growth path (BGP) exists. In a BGP, all variables grow at constant (but potentially different) rates; in particular, $g_{K,t} := \frac{K_{t+1}}{K_t}$ must be constant (i.e., capital must grow at a constant rate). From the law of motion of capital, we have

$$g_{K,t} = s \frac{Y_t}{K_t} + 1 - \delta,$$

thus, along a BGP, $y_t := \frac{Y_t}{K_t}$ must be constant. From the production function, we have

$$y_t = \left(\frac{A_t N_t}{K_t} (1 - s_R) \right)^{1-\alpha};$$

therefore, y_t is constant if and only if $\frac{A_t N_t}{K_t}$ is constant; that is,

$$\frac{A_t N_t}{K_t} = \frac{A_{t+1} N_{t+1}}{K_{t+1}} \iff g_K = \frac{A_{t+1}}{A_t} n.$$

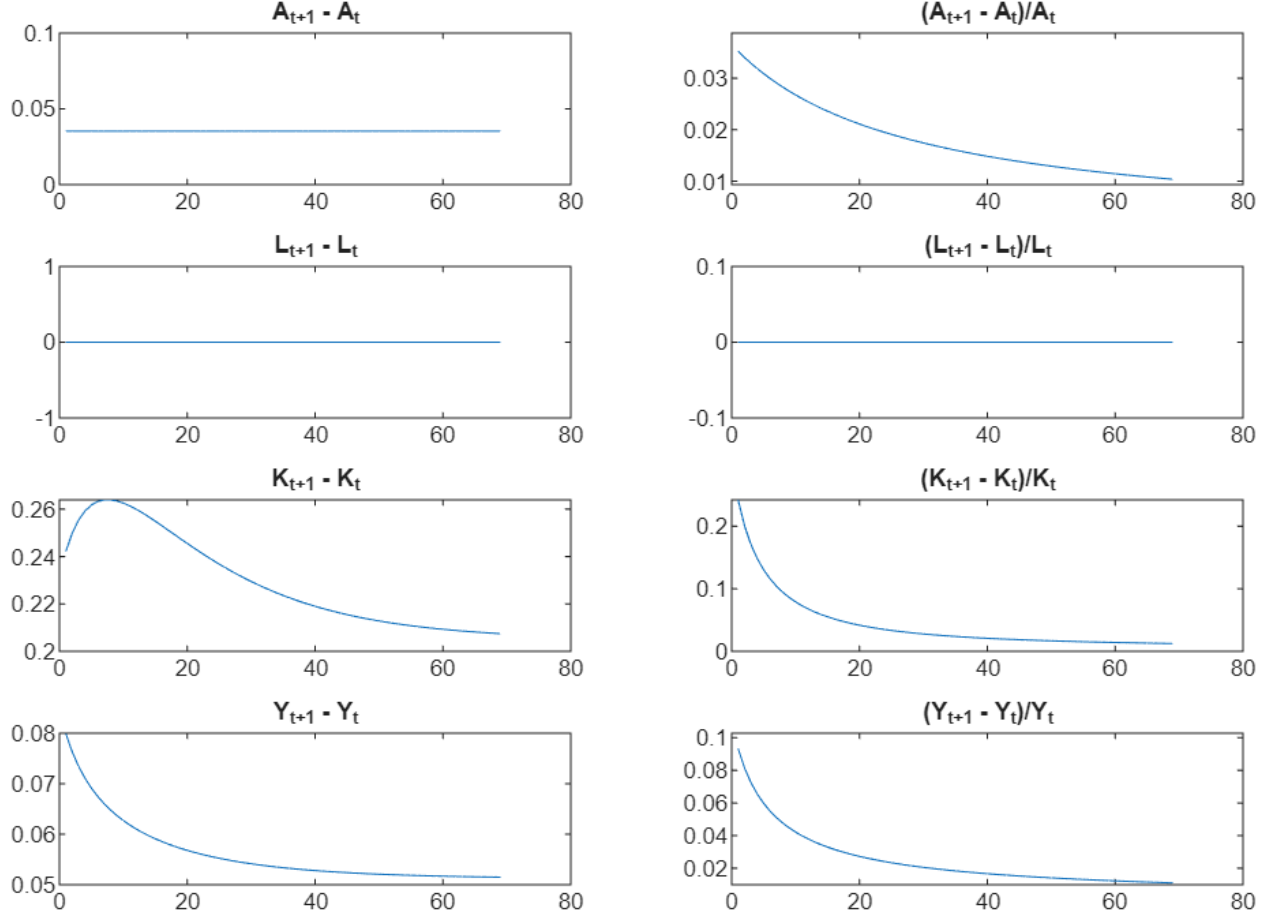
So, a BGP requires $g_{A,t} := \frac{A_{t+1}}{A_t} = \frac{g_K}{n}$ to be constant (that is, A_t must grow at a constant rate). From the production function of ideas, we see that $g_{A,t}$ being constant requires $N_t^\lambda, A_t^{\phi-1}$ to be constant; that is,

$$N_t^\lambda A_t^{\phi-1} = N_{t+1}^\lambda A_{t+1}^{\phi-1} \iff 1 = n^\lambda g_{A,t}^{\phi-1}.$$

If we rule out a degenerate BGP with eventually no population (that is, we rule out $n < 1$), and given that A_t cannot decrease over time (i.e., $g_A \geq 1$), then there cannot be a BGP if $\phi > 1$.¹

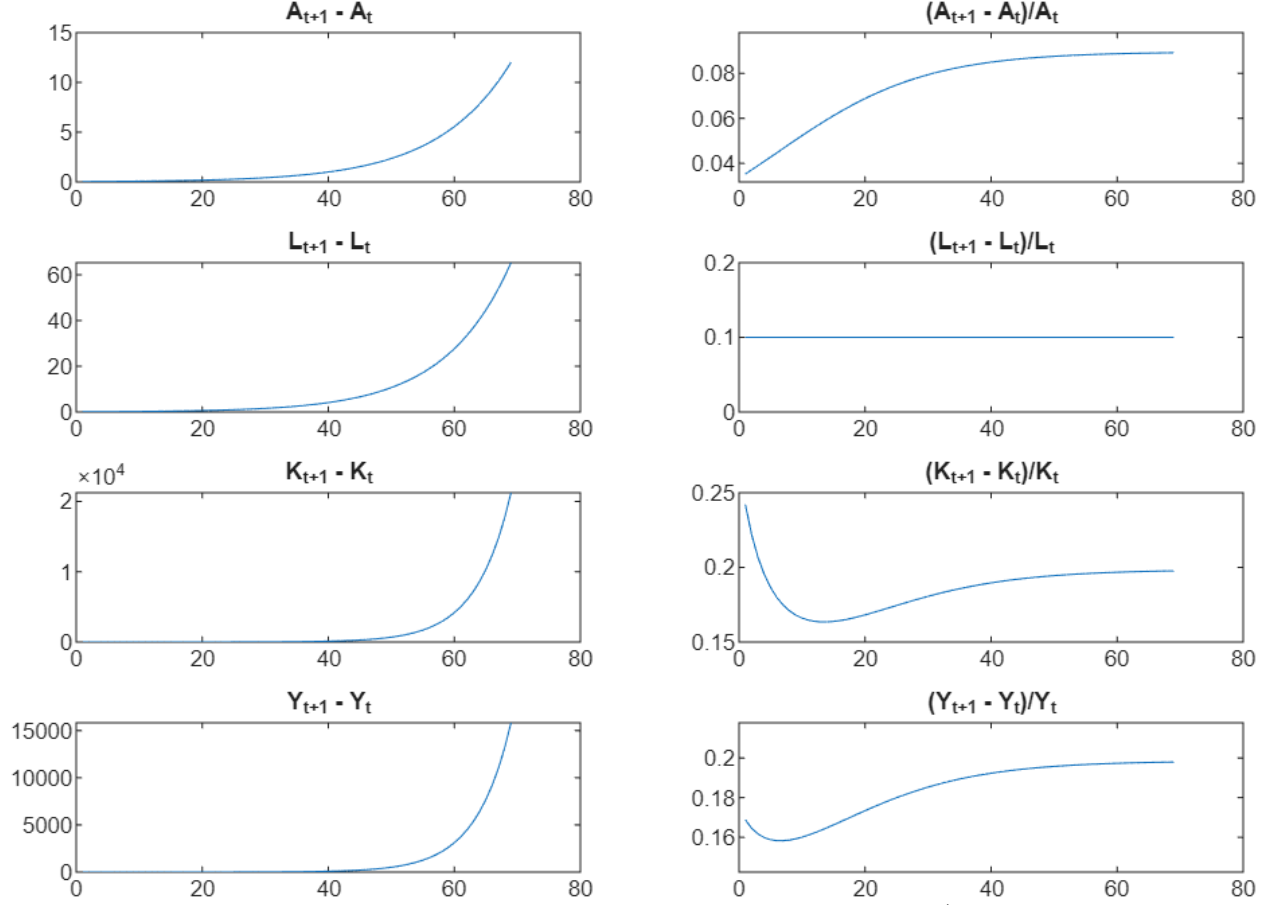
We have seen that if there is a BGP, it must be that $\phi \leq 1$. But does $\phi \leq 1$ imply that a BGP exists? In other words, we have shown that $\phi \leq 1$ is a necessary condition for the existence of a BGP, and now we ask whether it is also sufficient. In Appendix 2.1 (you are not expected to prove this; it is included for completeness), I show that if $\phi < 1$, then there exists a BGP and, in addition, the economy converges to it.

¹If there is positive population growth, we can also rule out the case $\phi = 1$, since this would imply the contradiction $1 = n^\lambda > 1$.

Figure 1: Convergence to BGP with $\phi = 0$ and no population growth

Notes. Simulation of the model with no knowledge spillovers ($\phi = 0$) and no population growth ($n = 1$). The remaining parameter values are $\alpha = 0.3$, $s_R = 0.2$, $s = 0.4$, $\delta = 0.1$, $\theta = 0.15$, and $\lambda = 0.9$. Initial conditions: $K_0 = 1$, $L_0 = 1$, $A_0 = 1$.

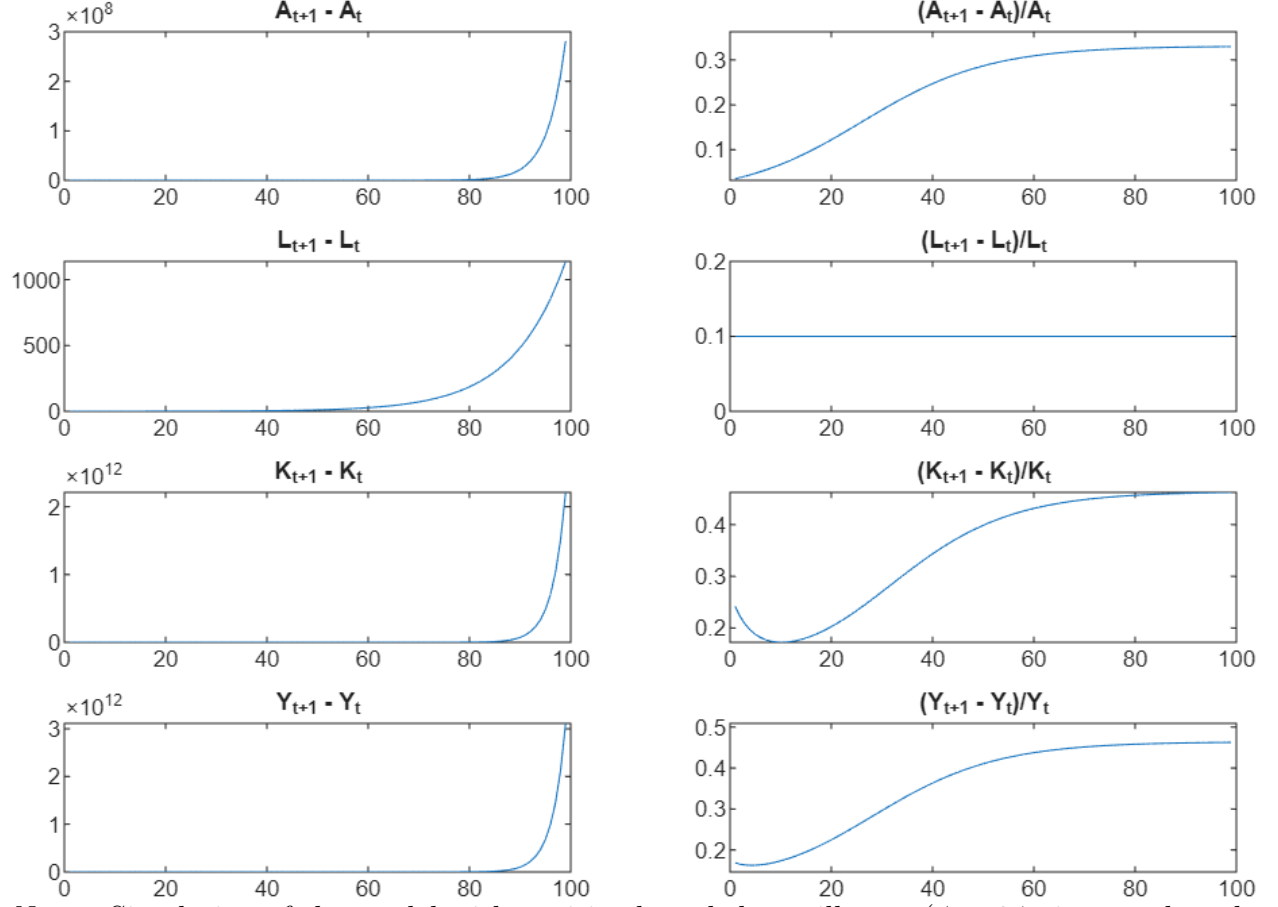
In the absence of population growth (so the number of researchers is constant) and with $\phi = 0$ (so the flow of new ideas is independent of the existing stock of knowledge), $A_{t+1} - A_t$ is constant. As A_t increases, however, its growth rate declines and converges to zero. Therefore, although technological progress continues indefinitely, its effect on growth becomes diminishing and eventually negligible. Sustained growth requires an increasing number of researchers, as illustrated in the next figure.

Figure 2: Convergence to BGP with $\phi = 0$ and with population growth

Notes. Simulation of the model with positive knowledge spillovers ($\phi = 0.7$, that is, weak scale effects $\phi < 1$) and with 10% population growth ($n = 1.1$). The remaining parameter values are, as in the previous figures, $\alpha = 0.3$, $s_R = 0.2$, $s = 0.4$, $\delta = 0.1$, $\theta = 0.15$, and $\lambda = 0.9$. Initial conditions: $K_0 = 1$, $L_0 = 1$, $A_0 = 1$.

Compared to the previous figure with no knowledge spillovers, we see that In the right panels, the relevant feature is the level to which the variables converge, as the behavior in the initial periods depends on the chosen initial conditions (for instance, the growth rate of A_t would be decreasing toward the BGP level if A_0 were sufficiently small). In contrast to the previous figure, L_t now grows at a constant rate. As a result, the flow of new ideas, $A_{t+1} - A_t$, increases without bound, and the growth rate of A_t converges to a positive constant. This, in turn, sustains a positive growth rate of output.

Figure 3: Convergence to BGP with positive knowledge spillovers and with population growth



Notes. Simulation of the model with positive knowledge spillovers ($\phi = 0.7$, i.e., weak scale effects with $\phi < 1$) and with 10% population growth ($n = 1.1$). The remaining parameter values are, as in the previous figures, $\alpha = 0.3$, $s_R = 0.2$, $s = 0.4$, $\delta = 0.1$, $\theta = 0.15$, and $\lambda = 0.9$. Initial conditions: $K_0 = 1$, $L_0 = 1$, $A_0 = 1$.

Compared to the previous figure with no knowledge spillovers, we now see that as A_t increases, it becomes easier to generate new ideas. In particular, $A_{t+1} - A_t$ increases much faster than in the previous figure, where its growth was driven solely by population growth (i.e., by the increase in the number of researchers).

2 Appendix

2.1 Proof of Convergence to BGP (Sufficiency)

First, the following Lemma guarantees that $\phi \leq 1$ implies that $g_{A,t}$ converges to $n^{\frac{\lambda}{1-\phi}}$:

Lemma 1 *If $\phi \leq 1$, then $g_{A,t}$ converges to $n^{\frac{\lambda}{1-\phi}}$.*

Proof. We are going to show that if $g_{A,t} > n^{\frac{\lambda}{1-\phi}}$, then $|g_{A,t} - n^{\frac{\lambda}{1-\phi}}| > |g_{A,t+1} - n^{\frac{\lambda}{1-\phi}}|$ (the proof that if $g_{A,t} < n^{\frac{\lambda}{1-\phi}}$, then $|g_{A,t} - n^{\frac{\lambda}{1-\phi}}| > |g_{A,t+1} - n^{\frac{\lambda}{1-\phi}}|$ is analogous):

First, we show that $g_{A,t} > n^{\frac{\lambda}{1-\phi}} \implies g_{A,t+1} \leq g_{A,t}$.

Indeed, from the definition of $g_{A,t+1}$, then: $g_{A,t+1} = 1 + \theta s_R^\lambda N_{t+1}^\lambda A_{t+1}^{\phi-1} = 1 + \theta s_R^\lambda (n N_t)^\lambda (g_{A,t} A_t)^{\phi-1} \leq 1 + \theta s_R^\lambda N_t^\lambda A_t^{\phi-1} = g_{A,t}$.

Second, we show that $g_{A,t} > n^{\frac{\lambda}{1-\phi}} \implies g_{A,t+1} \geq n^{\frac{\lambda}{1-\phi}}$: For this, note that $g_{A,t+1} = 1 + \theta s_R^\lambda N_t^\lambda A_t^{\phi-1} \frac{n^\lambda}{g_{A,t}^{1-\phi}} = 1 + (g_{A,t} - 1) \frac{n^\lambda}{g_{A,t}^{1-\phi}}$; so, consider $g_{A,t+1}$ as a function of $g_{A,t}$: $g_{A,t+1}(g_A) := 1 + (g_A - 1) \frac{n^\lambda}{g_A^{1-\phi}}$. Note that: (i) $g_{A,t+1}(n^{\frac{\lambda}{1-\phi}}) = n^{\frac{\lambda}{1-\phi}}$; and (ii) $g'_{A,t+1}(g_A) = \frac{n^\lambda}{g_A^{1-\phi}} - (1-\phi)(g_A - 1) \frac{n^\lambda}{g_A^{2-\phi}} = \frac{n^\lambda}{g_A^{2-\phi}} [\phi g_A + 1 - \phi] > 0$. So: $g_A > n^{\frac{\lambda}{1-\phi}} \implies g_{A,t+1}(g_A) > n^{\frac{\lambda}{1-\phi}}$.

So, combining the two results, we have: $|g_{A,t} - n^{\frac{\lambda}{1-\phi}}| = g_{A,t} - n^{\frac{\lambda}{1-\phi}} > g_{A,t+1} - n^{\frac{\lambda}{1-\phi}} = |g_{A,t+1} - n^{\frac{\lambda}{1-\phi}}|$, where the first inequality comes from $g_{A,t+1} < g_{A,t}$, and the following equality comes from $n^{\frac{\lambda}{1-\phi}} < g_{A,t+1}$. ■

Next, recall that if $g_{K,t} = g_{A,t} n$, then y_t is constant. Then, given that $g_{A,t} \rightarrow n^{\frac{\lambda}{1-\phi}}$; then we need to show that $g_{K,t} \rightarrow g_K^* := n^{1+\frac{\lambda}{1-\phi}}$; and from $g_{K,t} = s y_t + (1-\delta)$, this is equivalent to show that $y_t \rightarrow y^* := \frac{n^{1+\frac{\lambda}{1-\phi}} - (1-\delta)}{s}$.

Proof. We are going to show that if $y_t > y^*$, then $|y_t - y^*| > |y_{t+1} - y^*|$ (the proof that if $y_t < y^*$, then $|y_t - y^*| > |y_{t+1} - y^*|$ is analogous):

First, we show that for t sufficiently large, if $y_t > y^*$, then $y_t > y_{t+1}$: Note that $y_t > y^*$ implies $g_{K,t} = s y_t + (1-\delta) > s y^* + (1-\delta) = g_K^*$; and so: $y_{t+1} = \left(\frac{A_{t+1} N_{t+1}}{K_{t+1}} (1-s_R) \right)^{1-\alpha} = \left(\frac{A_t g_{A,t} N_t}{K_t g_{K,t}} (1-s_R) \right)^{1-\alpha} < \left(\frac{A_t N_t}{K_t} (1-s_R) \right)^{1-\alpha} = y_t$, where for the inequality I have used that t is sufficiently large, and so $g_{A,t} n$ is arbitrarily close to g_K^* (since we already know that $g_{A,t} n \rightarrow n^{1+\frac{\lambda}{1-\phi}} = g_K^*$).

Second, we show that for t sufficiently large, if $y_t > y^*$, then $y_{t+1} > y^*$: Note that $y_{t+1} = y_t \left(\frac{g_{A,t} n}{g_{K,t}} \right)^{1-\alpha} = y_t \left(\frac{g_{A,t} n}{s y_t + 1 - \delta} \right)^{1-\alpha}$; so, we can define the function $y_{t+1}(y) = y_t \left(\frac{g_K^*}{s y_t + 1 - \delta} \right)^{1-\alpha}$,

which satisfies: (i) $y_{t+1}(y^*) = y^* \left(\frac{g_K^*}{s y^* + 1 - \delta} \right)^{1-\alpha} = y^*$; and (ii)

$$y'_{t+1}(y) = \frac{g_K^{*1-\alpha}}{(s y_t + 1 - \delta)^{2-\alpha}} [s y + 1 - \delta - (1 - \alpha) s] > 0, \text{ for any } y \geq y^*,$$

where the inequality comes from $s y + 1 - \delta \geq s y^* + 1 - \delta = g_K^* \geq 1 > (1 - \alpha) s$. Therefore, for $y_t > y^*$, and t sufficiently large (so that y_{t+1} is arbitrarily close to $y_{t+1}(y_t)$), we have $y_{t+1} > y^*$.

So, combining the two results, we have: $|y_t - y^*| = y_t - y^* > y_{t+1} - y^* = |y_{t+1} - y^*|$, where the first inequality comes from $y_{t+1} < y_t$, and the following equality comes from $y_{t+1} \geq y^*$.

■

2.2 Expanding varieties model: endogenous researchers

Households and population growth: We assume a representative household that supplies inelastically N_t units of labor growing at a constant rate: $N_{t+1} = n N_t$. The representative household has CIES preferences:

$$U = \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\theta} - 1}{1-\theta}.$$

The household supplies inelastically N_t units of labor, which grow at a constant rate $N_{t+1} = n N_t$, starts with s_0 savings, and has the following budget constraint at time t :

$$(1 + r_t) s_{t-1} + w_t N_t = c_t + s_t.$$

Final good: The final good is produced with the following production function that uses labor L_t and the different varieties of intermediate goods available in the economy:

$$Y_t = \frac{1}{1-\alpha} \left(\int_0^{A_t} x_{j,t}^{1-\alpha} dj \right) L_t^\alpha,$$

where A_t is the amount of varieties available in period t , and $x_{j,t}$ the quantity of intermediate good j produced in period t .

Intermediate goods: There is a monopolist producing each variety, with a linear production function: we simply assume that each unit of intermediate good produced requires ν units of final good. To simplify expressions, we assume the normalization $\nu = 1 - \alpha$.

Research: The labor employed in research $L_{R,t}$ produces new varieties that add up to the existing ones, according to:

$$A_{t+1} - A_t = \psi L_{R,t}^\lambda A_t^\phi, \quad \lambda \in (0, 1), \phi < 1.$$

2.2.1 Equilibrium

Final good producer. The final good producer chooses labor L_t and intermediate goods $x_{j,t}$ to maximize profits:

$$\max_{x_{j,t}, L_t} \frac{1}{1-\alpha} \left(\int_0^{A_t} x_{j,t}^{1-\alpha} dj \right) L_t^\alpha - \int_0^{A_t} p_{j,t} x_{j,t} dj - w_t L_t.$$

From the FOCs, we get the demand functions for labor and intermediates:

$$w_t = \alpha \frac{Y_t}{L_t}, \quad p_{j,t} = x_{j,t}^{-\alpha} L_t^\alpha. \quad (1)$$

These conditions state that labor costs account for a fraction α of Y_t , and intermediate goods costs for the remaining fraction $1 - \alpha$ of Y_t ,² so, the final good producer earns zero profits.

Intermediate goods problem. Each monopolist chooses $x_{j,t}$ to maximize profits, internalizing the demand it faces:

$$\max_{x_{j,t}} p_{j,t} x_{j,t} - \nu x_{j,t}, \text{ s.t. } p_{j,t} = x_{j,t}^{-\alpha} L_t^\alpha$$

The FOC delivers the optimal price, which is a constant markup $\frac{1}{1-\alpha}$ over the marginal cost, which is ν :

$$\frac{\partial \pi_{j,t}}{\partial x_{j,t}} = \frac{\partial p_{j,t}}{\partial x_{j,t}} x_{j,t} + (p_{j,t} - \nu) = 0 \iff p_{j,t} = \frac{\nu}{1-\alpha} = 1.$$

And the amount $x_{j,t}$ demanded at this price is:

$$x_{j,t} = L_t.$$

Plugging this into the final good production function:

$$Y_t = \frac{1}{1-\alpha} A_t L_t.$$

²Indeed: $\int_0^{A_t} p_{j,t} x_{j,t} dj = \int_0^{A_t} x_{j,t}^{1-\alpha} L_t^\alpha dj = (1-\alpha) Y_t.$

So, Y_t grows at the same rate as $A_t L_t$. And substituting L_t using the FOC of L_t , we get that wages grow linearly with A_t :

$$w_t = \frac{\alpha}{1 - \alpha} A_t. \quad (2)$$

And the profits of each intermediate good monopolist:

$$\pi_{j,t} = (p_{j,t} - \nu) x_{j,t} = \alpha L_t.$$

Innovators problem. The innovators know that by employing $L_{R,t}$ units of labor in research, they will create $\Delta A = \psi L_{R,t}^\lambda A_t^\phi$ new varieties, and they will be able to sell a patent with the monopoly rights to produce this variety at its market price $V_{j,t+1}$, which is the present-value of all the future monopoly profits:³

$$V_{j,t} = \sum_{t'=t}^{\infty} \frac{\pi_{j,t'}}{\prod_{k=t+1}^{t'} (1 + r_k)}$$

To simplify this expression, we focus on a balanced growth path, where the interest rate is constant $r_t = r$, and the labor allocated to research is a constant share s_R (note that, in contrast to the toy model in Section 1.2, where the constant fraction of labor allocated to production is exogenous, here it arises endogenously from the choices of firms) of total labor $L_t = (1 - s_R) N_t$; and so, $L_{t+1} = (1 - s_R) N_{t+1} = (1 - s_R) N_t n = L_t n$; so: $\pi_{j,t'} = \pi_{j,t'-1} n$. Therefore, we can rewrite the value of variety j as:

$$\begin{aligned} V_{j,t} &= \sum_{t'=t}^{\infty} \frac{\pi_{j,t'}}{(1+r)^{t'-t}} = \pi_{j,t} + \frac{n}{1+r} \sum_{t'=t+1}^{\infty} \frac{\pi_{j,t'-1}}{(1+r)^{t'-1-t}} \\ &= \pi_{j,t} + \frac{n}{1+r} \sum_{t'=t}^{\infty} \frac{\pi_{j,t'}}{(1+r)^{t'-t}} = \pi_{j,t} + \frac{n}{1+r} V_{j,t} \end{aligned}$$

And solving for $V_{j,t}$, and substituting profits, we have:

$$V_{j,t} = \frac{1+r}{1+r-n} \alpha L_t \quad (3)$$

The innovators choose $L_{R,t}$ to maximize

$$\max_{L_{R,t}} \psi L_{R,t}^\lambda A_t^\phi \frac{V_{j,t+1}}{1+r_{t+1}} - w_t L_{R,t},$$

³We use the convention that $\prod_{k=t+1}^t (1+r_k)^{-1} = 1$.

The FOC is

$$\lambda \psi L_{R,t}^{\lambda-1} A_t^\phi \frac{V_{j,t+1}}{1+r_{t+1}} = w_t.$$

Substituting $w_t = \frac{\alpha}{1-\alpha} A_t$, and if we focus on the BGP, then $\frac{V_{j,t+1}}{1+r} = \frac{n\alpha}{1+r-n} L_t$, and $L_t = (1-s_R)N_t$ and $L_{R,t} = s_R N_t$, so we have:

$$\lambda \psi s_R^{\lambda-1} (1-s_R) N_t^\lambda A_t^{\phi-1} \frac{n(1-\alpha)}{1+r-n} = 1,$$

which defines implicitly s_R . Note that for s_R to be constant, as in the toy model, we require $N_t^\lambda A_t^{\phi-1}$ to be constant; and if this is the case then A_t also grows at a constant rate, which in turn implies from 2 that wages also grow at a constant rate (at the same rate). And from the aggregate resource constraint:

$$Y_t = C_t + \nu \int_0^{A_t} x_{j,t} dj,$$

substituting $x_{j,t} = L_t$ and $Y_t = \frac{1}{\alpha} A_t L_t$, and recall $\nu = 1 - \alpha$:

$$C_t = \frac{1}{1-\alpha} A_t L_t - (1-\alpha) A_t L_t = \left(\frac{1}{1-\alpha} - (1-\alpha) \right) A_t L_t.$$

And so, C_t grows at the same rate as Y_t . Therefore, there exists a BGP. The only remaining question is about convergence to it, which reduces to check whether $N_t^\lambda A_t^{\phi-1}$ converges to a constant, which we already saw in the toy model it is the case if $\phi < 1$. For completeness, I finally show the solution of the consumer problem, which given the growth rate of the economy in the BGP, pins down the interest rate (which we have used to determine the value of a successful innovation, which determines the equilibrium fraction of labor allocated to research):

Household problem. The household chooses consumption at each period to maximize life-time utility subject to the budget constraint.⁴ Similarly to what we did in the OLG model, we can convert this problem into an unconstrained one: From the budget constraint in each period, we get consumption as a function of previous and current saving $c(s_{t-1}, s_t) := w_t N_t - s_t + (1+r_t)s_{t-1}$. Substituting C_t in the life-time utility U , the household problem becomes:

$$\max_{s_t} \sum_{t=0}^{\infty} \beta^t \frac{c(s_{t-1}, s_t)^{1-\theta} - 1}{1-\theta}.$$

⁴And that it cannot run a Ponzi scheme, where it sustainedly pays previous debt issuing more debt.

The FOC with respect to s_t is:

$$\beta^t C_t^{-\theta} \frac{\partial c(s_{t-1}, s_t)}{\partial s_t} + \beta^{t+1} C_{t+1}^{-\theta} \frac{\partial c(s_t, s_{t+1})}{\partial s_t} = 0.$$

And using that $\frac{\partial c(s_{t-1}, s_t)}{\partial s_t} = -1$ and $\frac{\partial c(s_t, s_{t+1})}{\partial s_t} = (1 + r_{t+1})$, we have the same Euler equation we obtained in the OLG model with CIES preferences:

$$\beta^t C_t^{-\theta} = \beta^{t+1} C_{t+1}^{-\theta} (1 + r_{t+1}) \implies \frac{C_{t+1}}{C_t} = (\beta(1 + r_{t+1}))^{\frac{1}{\theta}}.$$

So, in the BGP, letting $g := \frac{C_{t+1}}{C_t}$:

$$r = \frac{g^\theta}{\beta} - 1. \tag{4}$$